Requirements Gathering Specification

Project: TeleScope - Customer Intelligence & Recommendation Platform

1. Identify Stakeholders

Internal Team: Data Engineers, Machine Learning Engineers, Backend Developers, Frontend Developers (Streamlit), and Project Managers.

External Stakeholders: Telecom customers in Egypt, Telecommunications Operators (Vodafone Egypt, Orange Egypt, WE, e& Egypt), and Academic Evaluators.

Stakeholder Mapping: Stakeholders are categorized by influence and interest to proactively manage alignment, privacy compliance, and performance expectations.

2. Plan the Approach

Project Objectives: Construct an automated, data-driven system that aggregates customer reviews, structured survey responses, and live telecom tariff plans to deliver personalized, explainable operator rankings.

High-Level Scope: Data collection from Google Play and surveys; scalable object storage via Microsoft Azure Blob Storage; data processing using PySpark; PostgreSQL data warehousing; and interactive serving via the Streamlit UI.

Elicitation Techniques: One-on-one stakeholder discussions, automated app review scraping, user feedback surveys, and technical interface mapping.

3. Conduct Elicitation

Business Needs Focus: Elicitation centers on solving core customer pain points such as network quality, price-to-value ratios, bundle validity, and customer service, rather than premature tool constraints.

Multi-Channel Ingestion: Harvesting customer reviews from Google Play Store, quantitative questionnaire data, and public operator tariff APIs/scrapers.

4. Document Observations and Findings

Category	Requirement Specification
Functional	• Automated ingestion of Google Play reviews and offer data via Python/Airflow. • Storage of raw unstructured telemetry in the Microsoft Azure Blob Storage data lake. • Data cleaning & processing with PySpark. • Machine Learning execution (Sentiment Analysis and the Hybrid Recommender: rule-based filters, then ML ranking). • Interactive UI in Streamlit backed by a Python services layer.
Non-Functional	• High performance and low-latency recommendation inference (under 2 s). • Scalable distributed storage and processing architecture. • Robust data privacy compliance and clean pipeline fault tolerance.
Constraints	• Reliance on publicly accessible app reviews and structured survey inputs. • Storage on Microsoft Azure Blob Storage; subscription access and cost must be confirmed.

Recommendation method. Hybrid recommender: Step 1 applies rule-based filters for hard constraints (budget, quota, validity); Step 2 applies ML ranking to the eligible offers using validated features.

5. Analyze and Prioritize (MoSCoW Framework)

Priority Level	Capabilities & Features Included
Must Have	Python data scrapers, Azure Blob Storage, Airflow orchestration, PySpark cleaning, Sentiment Analysis model, Hybrid Recommender (rule-based filters, then ML ranking), and Streamlit frontend with a Python services layer.
Should Have	PostgreSQL data warehouse integration and Power BI reporting dashboards for management.
Could Have	No items assigned at this stage.
Won't Have	Direct operator billing integration or carrier account management in Phase 1.

6. Verify and Validate

Stakeholder Review: Structured architecture walkthroughs and Streamlit UI prototype reviews conducted with project stakeholders and evaluators.

Formal Sign-off: Formal approval secured on pipeline stages, tech stack selection (Azure Blob Storage, PySpark, Streamlit), and core data sources prior to full engineering deployment.

7. Manage Changes

Traceability Matrix: Comprehensive baseline mapping linking raw data requirements to processing DAGs, ML models, and Streamlit interface components.

Change Control Process: Defined change control workflow to handle evolving telecom tariff structures, new operator offerings, or mid-project data pipeline discoveries.